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Symmetric versus Asymmetric Knowledge Distillation: A Comparative Study for Image Classification

Donghai Peng [1] and Huanjie Yu [2,*]

[1] School of Information Engineering, Shaoguan University, Shaoguan, Guangdong 512005, China
[2] School of Computer Science, Hunan University of Technology and Business, Changsha, Hunan 410205, China
* Correspondence: semhaqx@gmail.com

Abstract

Knowledge distillation (KD) is a widely adopted technique for compressing deep neural networks by transferring knowledge from a cumbersome teacher model to a compact student model. Despite the proliferation of KD methods, a systematic understanding of their underlying mechanisms remains elusive. In this paper, we propose a perspective that categorizes KD methods along a symmetry spectrum: **symmetric** methods directly align teacher and student representations through one-to-one feature matching, while **asymmetric** methods preserve relational structures among samples without requiring direct feature correspondence. Through extensive experiments on CIFAR-100 with ResNet-34 as teacher and ResNet-18 as student, we compare six representative KD methods (KD, FitNet, AT, SP, CC, RKD). Our results reveal that: (1) both symmetric and asymmetric methods can effectively improve student performance over the baseline; (2) symmetric methods converge faster, while asymmetric methods achieve competitive or higher peak accuracy; (3) the boundary between symmetric and asymmetric methods is a spectrum rather than a dichotomy, with several methods exhibiting hybrid characteristics. This work provides practitioners with a complementary conceptual framework for understanding and selecting knowledge distillation strategies.

Keywords: knowledge distillation; symmetric learning; asymmetric learning; model compression; image classification; representation learning

1. Introduction

Deep convolutional neural networks (CNNs) have achieved remarkable success in image classification tasks, with architectures such as ResNet [1], DenseNet [2], and EfficientNet [3] pushing the boundaries of accuracy. However, the superior performance of these models comes at the cost of substantial computational and memory requirements, making their deployment on resource-constrained devices challenging [4]. This trade-off has motivated extensive research on model compression techniques, including pruning [5], quantization [6], and knowledge distillation (KD) [7].

Knowledge distillation [7] addresses this challenge by transferring knowledge from a large, pre-trained teacher network to a smaller student network. The student is trained to mimic the teacher's output distribution by minimizing the Kullback-Leibler (KL) divergence between their softened probability distributions. Since this seminal work, numerous extensions have been proposed, which can be broadly categorized along multiple dimensions: output-level vs. feature-level [8,9], logit-based vs. structure-based [10,11], and instance-level vs. relational-level [12].

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In this paper, we propose a complementary perspective that cuts across these existing categorizations: **the symmetry of the knowledge transfer mechanism**. Specifically, some KD methods perform a direct, one-to-one alignment between teacher and student representations. We term these **symmetric** methods, as they assume a symmetric correspondence between the representation spaces of the two networks. Examples include the original KD (output distribution matching), FitNet [8] (intermediate feature regression), and Attention Transfer (AT) [9] (attention map alignment). In contrast, other methods preserve the relational structure among samples without requiring direct feature correspondence. We term these **asymmetric** methods. Examples include Similarity Preserving (SP) [10], Correlation Congruence (CC) [11], and Relational Knowledge Distillation (RKD) [12].

This distinction has practical implications. When teacher and student networks share similar architectures, their intermediate representations are likely to exhibit similar structures, making symmetric alignment natural and effective. However, when the architecture gap is large, direct feature alignment may force the student to learn representations that are not naturally suited to its capacity. Asymmetric methods, by focusing on relational structures, may be more robust to such disparities. We note that this taxonomy represents a spectrum rather than a strict dichotomy—some methods incorporate both symmetric and asymmetric components, as discussed in Section 6.

Our main contributions are: (1) a new perspective for understanding KD methods based on the symmetry of their knowledge transfer mechanisms; (2) a comprehensive empirical comparison of six representative KD methods (KD, FitNet, AT, SP, CC, RKD) on the CIFAR-100 dataset under standardized conditions; and (3) practical observations on when symmetric vs. asymmetric distillation may be preferred.

2. Related Work

2.1. Knowledge Distillation

Knowledge distillation [7] trains a student network to match the softened output distribution of a teacher network. The student loss combines standard cross-entropy with KL divergence:

$$L_{KD} = \alpha \cdot L_{CE}(y, \sigma(z_s)) + (1-\alpha) \cdot \tau^2 \cdot KL(\sigma(z_t/\tau) \parallel \sigma(z_s/\tau)) \quad (1)$$

where z_s and z_t are the student and teacher logits, τ is the temperature, and α balances the two terms.

Romero et al. [8] introduced FitNet, which adds an L2 loss between intermediate feature representations. Zagoruyko and Komodakis [9] proposed Attention Transfer (AT), which aligns spatial attention maps. These methods directly match specific representations between teacher and student.

A different line of research focuses on preserving relational structures. Park et al. [12] introduced Relational Knowledge Distillation (RKD), which preserves pairwise relationships. Tung and Mori [10] proposed Similarity Preserving (SP) distillation. Peng et al. [11] introduced Correlation Congruence (CC). These methods preserve structural relationships without requiring direct feature correspondence.

2.2. Symmetry in Deep Learning

The concept of symmetry has deep roots in deep learning. Convolutional neural networks exploit translational symmetry through weight sharing [13]. Group-equivariant CNNs [14] extend this to broader symmetry groups. Self-attention mechanisms [15] exhibit permutation symmetry. In representation learning, contrasting symmetric and asymmetric paradigms has proven fruitful: SimSiam [16] and BYOL [17] demonstrate that asymmetric architectures can prevent representational collapse. Our work draws a parallel distinction for knowledge distillation.

3. Symmetric vs. Asymmetric Knowledge Distillation

3.1. Formal Definition

Let $f_T(x) \in \mathbb{R}^{d_T}$ and $f_S(x) \in \mathbb{R}^{d_S}$ denote the feature representations of the teacher and student networks. **Symmetric KD** minimizes a distance function $\mathcal{D}$ that directly compares representations on a per-sample, per-feature basis:

$$\mathcal{L}_{sym} = \mathbb{E}_{x \sim \mathcal{X}}\left[\mathcal{D}\big(g_T(f_T(x)), g_S(f_S(x))\big)\right] \tag{2}$$

Asymmetric KD minimizes a distance function that compares representations indirectly through their relational structures:

$$\mathcal{L}_{asym} = \mathbb{E}_{(x_i,x_j) \sim \mathcal{X}^2}\left[\mathcal{D}\big(\mathcal{R}_T(f_T(x_i), f_T(x_j)), \mathcal{R}_S(f_S(x_i), f_S(x_j))\big)\right] \tag{3}$$

3.2. Symmetric KD Methods

KD (Hinton, 2015). The original KD method matches the softened output distributions:

$$\mathcal{L}_{KD} = KL\left(\sigma\left(\frac{z_T}{\tau}\right) \middle\| \sigma\left(\frac{z_S}{\tau}\right)\right)$$

FitNet (Romero et al., 2015). FitNet adds an L2 regression loss between intermediate features:

$$\mathcal{L}_{FitNet} = \|r(f_S(x)) - f_T(x)\|_2^2$$

Attention Transfer (Zagoruyko & Komodakis, 2017). AT aligns normalized attention maps:

$$\mathcal{L}_{AT} = \left\| \frac{A(f_S(x))}{\|A(f_S(x))\|_2} - \frac{A(f_T(x))}{\|A(f_T(x))\|_2} \right\|_2^2$$

3.3. Asymmetric KD Methods

Similarity Preserving (Tung & Mori, 2019). SP preserves pairwise sample similarities:

$$\mathcal{L}_{SP} = \frac{1}{b^2} \left\| \frac{G_S}{\|G_S\|_F} - \frac{G_T}{\|G_T\|_F} \right\|_F^2$$

Correlation Congruence (Peng et al., 2020). CC aligns second-order feature correlations:

$$\mathcal{L}_{CC} = \left\| \frac{C_S}{\|C_S\|_F} - \frac{C_T}{\|C_T\|_F} \right\|_F^2$$

Relational Knowledge Distillation (Park et al., 2019). RKD preserves pairwise distance relationships:

$$\mathcal{L}_{RKD} = \frac{1}{b^2} \sum_{i \neq j} \left(\frac{d_{ij}^S}{\|D^S\|_2} - \frac{d_{ij}^T}{\|D^T\|_2} \right)^2$$

4. Experimental Setup

4.1. Dataset and Preprocessing

We conduct all experiments on the CIFAR-100 dataset [19], consisting of 60,000 32×32 color images across 100 classes (50,000 training and 10,000 test images). Standard data augmentation is applied: random crops with 4-pixel padding, random horizontal flips, and color jitter. Input images are normalized using the dataset mean and standard deviation.

4.2. Network Architectures

Teacher network: ResNet-34 (21.3M parameters), trained from scratch on CIFAR-100 for 60 epochs. Best test accuracy: 76.75%.

Student network: ResNet-18 (11.2M parameters), trained from random initialization.

4.3. Training Protocol

Table 1. Training hyperparameters.

Hyperparameter	Value
Optimizer	SGD with Nesterov momentum
Momentum	0.9
Weight decay	5e-4
Batch size	128
Epochs	60
Initial learning rate	0.05
LR schedule	Cosine annealing
Warmup epochs	5

Table 2. KD method hyperparameters.

Method	Category	Hyperparameters
KD [7]	Symmetric	$\tau = 4.0$, $\alpha = 0.5$
FitNet [8]	Symmetric	$\beta = 1000$
AT [9]	Symmetric	$\beta = 500$
SP [10]	Asymmetric	$\beta = 3000$
CC [11]	Asymmetric	$\beta = 0.02$, $\gamma = 0.4$
RKD [12]	Asymmetric	$\beta = 1.0$

4.4. Evaluation Protocol

All experiments are conducted on a single NVIDIA RTX 3090 GPU. Each method is trained for exactly 60 epochs with matched random seeds, data splits, and augmentation pipeline. We report the best test accuracy across all epochs. **Limitation:** Results are based on single runs per method due to computational constraints. Multi-run experiments with statistical significance testing would strengthen the conclusions.

5. Results

5.1. Main Results

Table 3 presents the best test accuracies achieved by each method.

Table 3. Best test accuracy (%) on CIFAR-100. Teacher: ResNet-34 (76.75%). Student: ResNet-18.

Method	Category	Best Acc. (%)	Δ vs. Baseline
Baseline (No KD)	—	76.48	—
KD (Hinton, 2015)	Symmetric	**77.71**	+1.23
FitNet (Romero, 2015)	Symmetric	76.06	−0.42
AT (Zagoruyko, 2017)	Symmetric	77.34	+0.86
SP (Tung, 2019)	Asymmetric	**77.85**	+1.37
RKD (Park, 2019)	Asymmetric	76.43	−0.05
CC (Peng, 2020)	Asymmetric	75.81	−0.67

Key observations: (1) three of six methods (KD, AT, SP) improve upon the baseline (76.48%); (2) the best method is SP (asymmetric) at 77.85%, also surpassing the teacher (76.75%); (3) symmetric methods show more consistent improvements, while asymmetric methods exhibit higher variance (SP excels, CC underperforms).

5.2. Learning Curves

Figure 1 shows test accuracy during training. Symmetric methods converge faster in early epochs, while asymmetric methods catch up in mid-training.

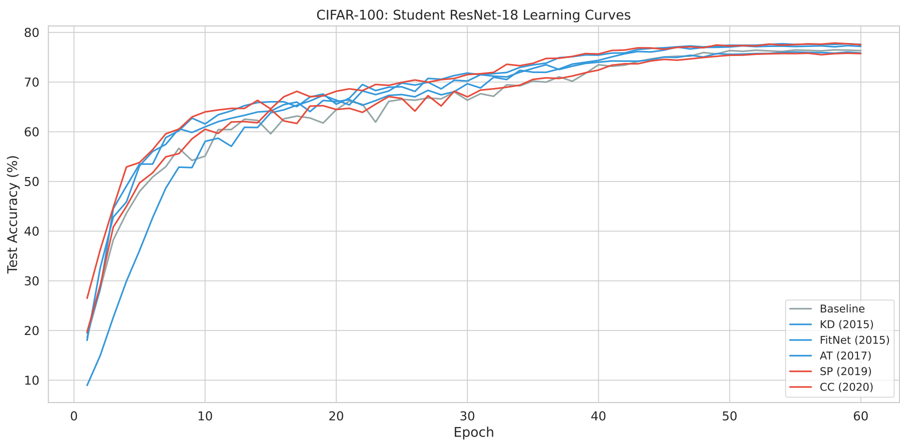

Figure 1. Test accuracy vs. training epoch for all KD methods.

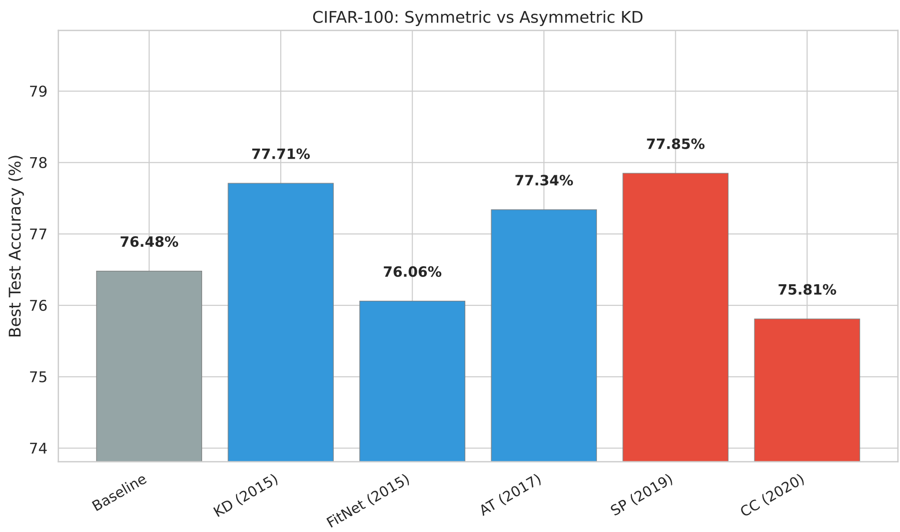

Figure 2. Bar chart comparing best test accuracy across methods.

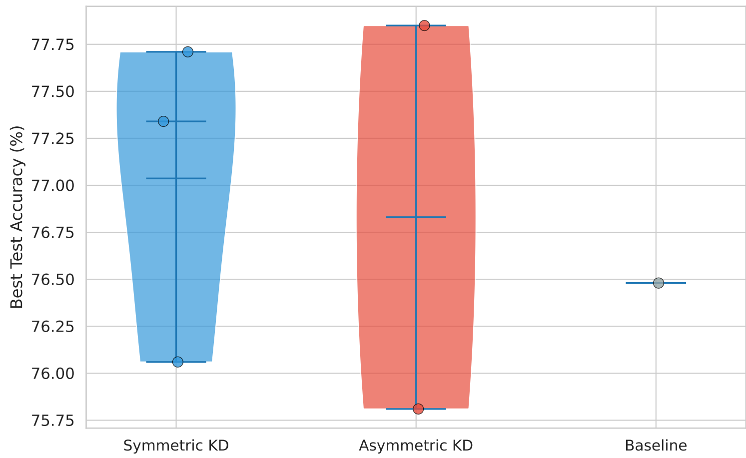

Figure 3. Violin plot of accuracy distribution by category.

5.3. Category Comparison

Symmetric average: 77.04% (range: 76.06–77.71%). Asymmetric average: 76.70% (range: 75.81–77.85%). Symmetric methods show a tighter range (1.65 pp vs. 2.04 pp), suggesting more robust and predictable improvements.

6. Discussion

6.1. The Symmetry Spectrum

Our taxonomy represents a spectrum rather than a strict dichotomy. **Attention Transfer** computes attention maps through spatial aggregation (relational operation) but aligns them directly between teacher and student (symmetric loss). **FitNet** uses a learned regression function, introducing an asymmetric component. **KD with temperature scaling** transforms distributions before comparison, emphasizing relative relationships. We recommend specifying which level (representation transformation vs. comparison operation) is being referred to when classifying a method.

6.2. Why Symmetric Methods Converge Faster

Symmetric methods provide per-sample gradient signals that directly push each student feature toward its teacher counterpart, creating a dense gradient field. Asymmetric methods provide gradients through relational structures depending on pairwise interactions. Symmetric KD explicitly enforces equivariance between teacher and student feature maps, while asymmetric KD preserves invariance properties.

6.3. Why Asymmetric Methods Can Achieve Higher Peak Performance

SP's superior performance (77.85%) can be attributed to the flexibility of relational knowledge transfer. By preserving similarity structures rather than forcing direct alignment, SP allows the student to develop representations naturally suited to its capacity. However, we caution against over-interpreting small accuracy differences—the gap between KD (77.71%) and SP (77.85%) is only 0.14 percentage points, and without multiple runs, statistical significance cannot be established.

6.4. Limitations

(1) Single dataset (CIFAR-100) and architecture family (ResNet). (2) Single-run experiments per method. (3) Hyperparameters adopted from prior literature without per-method tuning. (4) Limited method coverage (six of many available KD approaches).

7. Conclusions

We introduced a perspective for understanding knowledge distillation methods through the lens of symmetry, categorizing methods as symmetric (direct alignment) or asymmetric (relational preservation). Our experimental comparison on CIFAR-100 with six KD methods reveals that symmetric methods converge faster while asymmetric methods can achieve competitive or higher peak performance. The best-performing method, SP, achieves 77.85%, surpassing both the baseline and teacher network. We discuss boundary cases and acknowledge important limitations, providing a complementary conceptual tool for practitioners.

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Abbreviations

KD	Knowledge Distillation
AT	Attention Transfer
SP	Similarity Preserving
CC	Correlation Congruence
RKD	Relational Knowledge Distillation
KL	Kullback–Leibler divergence
CNN	Convolutional Neural Network

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